

a System and measurement boundaries



analog vector board



gripper context



power monitor

[TO BE MEASURED]

Instrumented hook-load grasping setup

Output: deployment, diagnostic, motor, power, force, and motion boundaries.

c Same-trial hook-load reflex



action context

[TO BE MEASURED]

Same-trial hook-load reflex frames

Output: approach, contact stop, hook load, bounded grip response, and recovery.

e Task performance and Vz-only ablation

[TO BE MEASURED]

Task performance and Vz-only causal ablation

Output: raw trials, intervals, fair Vz-only trigger, latency, and failures.

b Frozen two-event task protocol



grasping context

[TO BE MEASURED]

Frozen two-event grasp protocol

Output: CONTACT and hook-load event cycles with frozen control timing.

d Tactile-event-action dynamics

[TO BE MEASURED]

Synchronized tactile-event-action time series

Output: force, vectors, events, ADC/MCU state, commands, and slip on one clock.

f A1-A4 energy and break-even audit

[TO BE MEASURED]

Matched A1-A4 energy and break-even audit

Output: conversions, measured duty, rail energy, E_{task} , and event-rate break-even.